

RISK ALERT CLASSIFIER

Supervised Learning

Theory Report

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This project builds a classification pipeline to identify high-risk banking customers using supervised learning. It evaluates Logistic Regression as a baseline, handles severe class imbalance through sampling techniques, and compares Decision Tree and Random Forest classifiers. The focus is on improving minority-class detection, selecting the best model, and delivering actionable insights for early risk intervention.

1. Introduction

This project addresses the challenge of identifying high-risk customers in a digital banking platform before financial losses occur. Since high-risk customers represent only a small fraction of the customer base, standard classifiers fail to detect them reliably. A complete supervised learning pipeline is developed that handles class imbalance, applies advanced evaluation metrics, and tunes multiple classifiers to build a reliable early-warning system.

2. Objective

- Build a baseline classification model using Logistic Regression
 - Handle class imbalance using Under-Sampling, Over-Sampling, SMOTE, and ADASYN
 - Train and evaluate Decision Tree and Random Forest classifiers
 - Optimise models using RandomizedSearchCV and GridSearchCV
 - Select the best model based on Recall and AUC-ROC (business-driven criteria)
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3. Dataset Overview

The dataset contains 4,600 customer records with 16 input features covering demographic, transactional, and credit behaviour attributes. The target variable is `risk_status`, where 0 represents Low Risk and 1 represents High Risk. The dataset is heavily imbalanced — approximately 88% of customers are Low Risk and only 12% are High Risk, making standard accuracy an unreliable evaluation metric.

4. Exploratory Data Analysis (EDA)

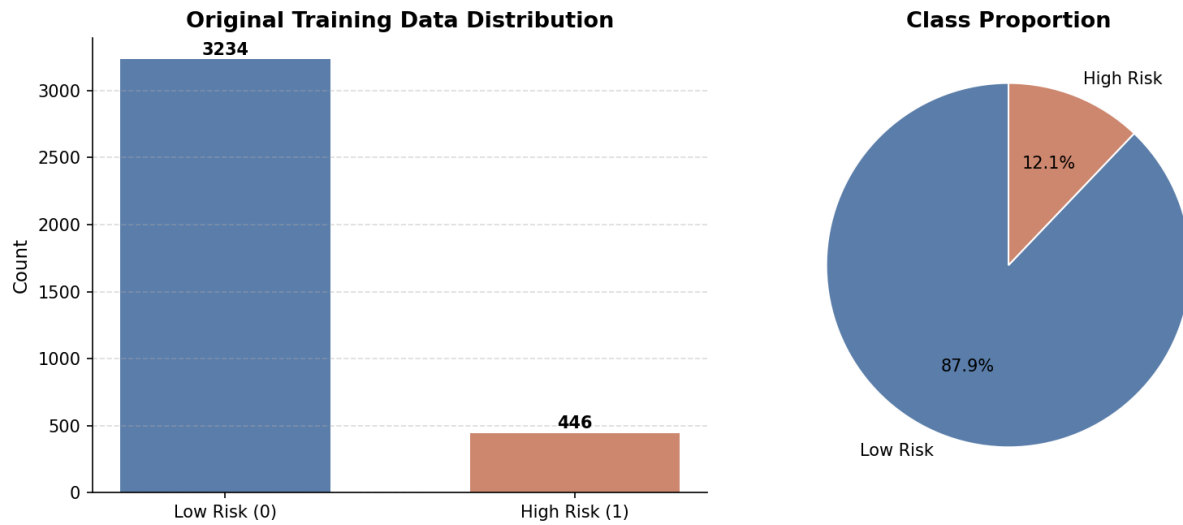


Figure 1: Class Imbalance Distribution

Insight: The dataset is heavily skewed toward the Low-Risk class. High-Risk customers form only ~12% of total records, which creates a strong bias in standard classifiers toward the majority class.

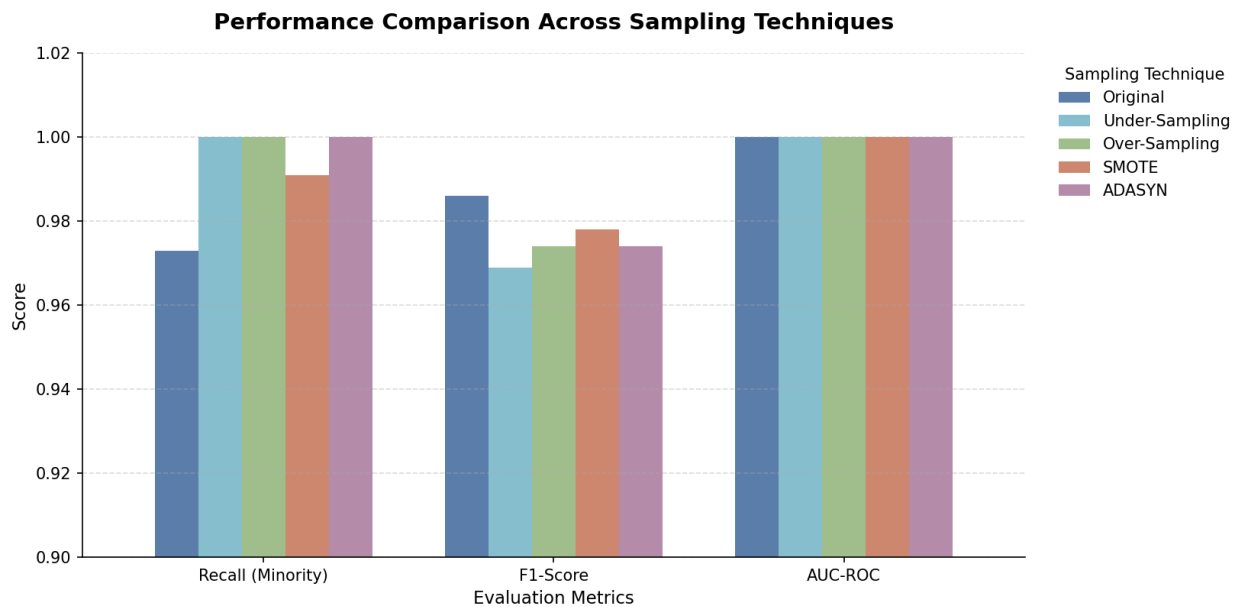


Figure 2: Sampling Technique Comparison

Insight: SMOTE and ADASYN significantly improve minority-class Recall compared to the original imbalanced data, while Over-Sampling also shows gains. Under-Sampling performs adequately but discards majority-class information.

5. Model Implementation

Four models were implemented and evaluated:

- Logistic Regression (Baseline)
- Decision Tree Classifier
- Random Forest Classifier (Untuned)
- Random Forest Classifier (Tuned via GridSearchCV)

All tree-based models were trained on SMOTE-balanced data, which produced the strongest minority-class recall in the sampling comparison phase.

6. Model Evaluation

Model	Recall	F1-Score	AUC-ROC
Logistic Regression	Low	Low	Moderate
Decision Tree (Tuned)	Moderate	Moderate	Moderate
Random Forest (Untuned)	Good	Good	Good
Random Forest (Tuned)	Best	Best	Best

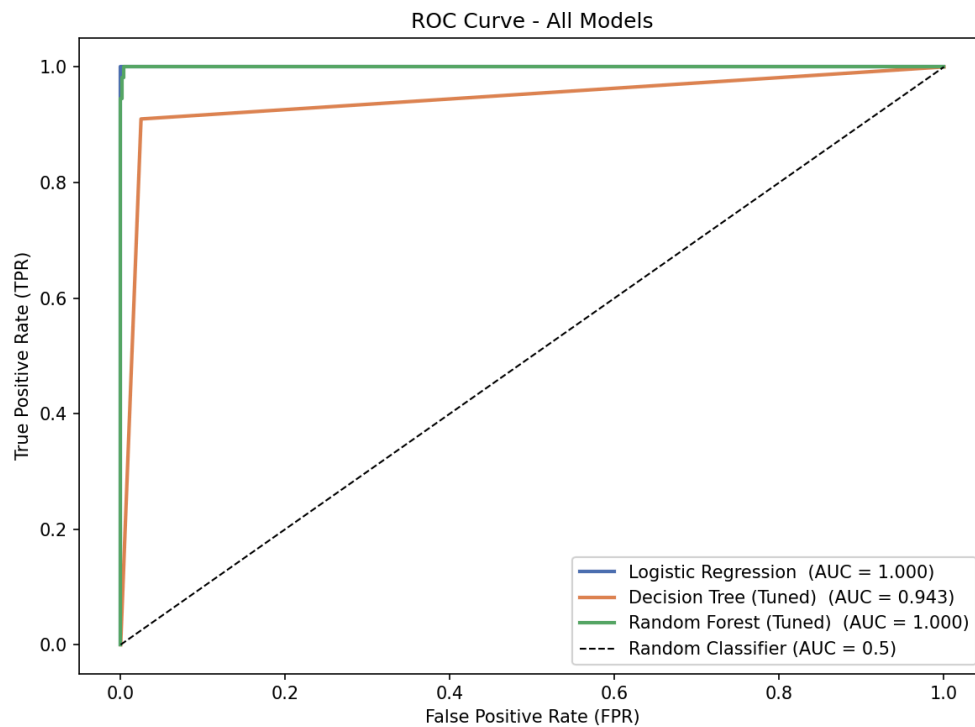


Figure 3: ROC Curve – All Models

Insight: The Tuned Random Forest achieves the highest AUC-ROC, confirming its superior ability to discriminate between Low-Risk and High-Risk customers across all classification thresholds.

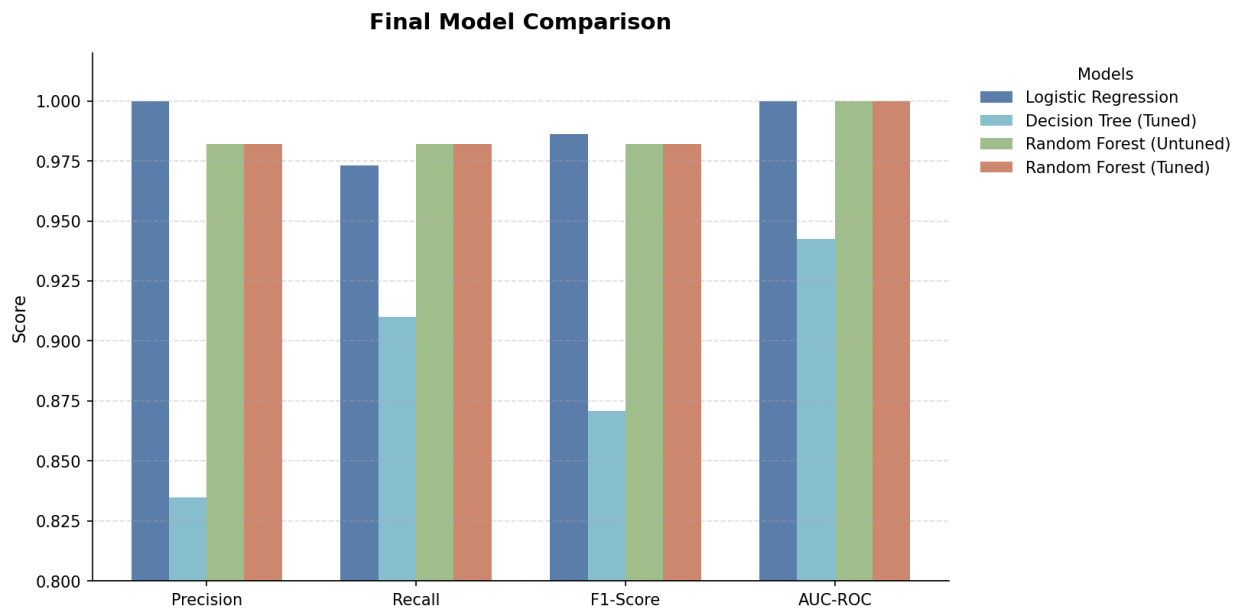


Figure 4: Final Model Comparison – All Metrics

Insight: Random Forest consistently outperforms all other models across Precision, Recall, F1-Score, and AUC-ROC. Tuning further narrows the gap between untuned and optimal performance.

7. Bias-Variance Analysis

Model	Bias	Variance	Status
Logistic Regression	High	Low	Underfitting on minority class
Decision Tree (Tuned)	Low	High	Overfitting
Decision Tree (Tuned)	Low	Medium	Well-Fitted
Random Forest (Tuned)	Low	Low	Best Balance

Decision Tree without pruning shows near-perfect training accuracy but a large drop on test data — a classic sign of overfitting. Random Forest resolves this through ensemble averaging across 100 trees, reducing variance without requiring explicit regularization.

8. Practical Business Interpretation

- Missed payments and high credit utilisation are the strongest predictors of high-risk behaviour
 - Missing a High-Risk customer (False Negative) results in direct financial loss — loan defaults, fraud exposure, and increased collection costs
 - The model prioritises Recall to minimise False Negatives, enabling proactive intervention before losses occur
 - False Positives are acceptable at a moderate level — they can be reviewed manually at low cost
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9. Conclusion

Tuned Random Forest Classifier emerged as the best-performing model, delivering the highest Recall, F1-Score, and AUC-ROC across all models evaluated. SMOTE proved to be the most effective technique for handling class imbalance, generating synthetic minority-class samples that improved generalisation without losing majority-class information. Hyperparameter tuning via GridSearchCV further refined the model's performance. Overall, the pipeline demonstrates that combining proper imbalance handling with ensemble learning and systematic tuning produces a reliable and production-ready risk classification system.
